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COURSE: COMPUTER NETWORK

COURSE CODE: CS-271

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LAB 07

Lab – 07

Vlan & Inter-Vlan Configuration

Warm up Task [30 Minutes]

- Answer each question in your own words (3 to 4 lines)

1. What is VLAN and its ranges

A VLAN (Virtual Local Area Network) is a logical grouping of devices on a network, regardless of their physical location, to improve security and efficiency. VLANs help segment traffic and reduce broadcast domains. VLAN IDs range from **1 to 4094**, where **1 and 4095 are reserved**, and **2–1001** are standard VLANs while **1006–4094** are extended.

2. Difference between VLAN and Inter-VLAN

A VLAN separates devices into different broadcast domains within the same switch. **Inter-VLAN** refers to communication between different VLANs, which requires a Layer 3 device like a router or Layer 3 switch to route traffic between them.

2. Pros and Cons of VLAN and Inter-VLAN

VLAN Pros:

- Enhances security by isolating traffic
- Reduces broadcast traffic

VLAN Cons:

- Requires careful planning and configuration
- Misconfigurations can cause connectivity issues

Inter-VLAN Pros:

- Enables communication between VLANs
- Centralized routing improves control

Inter-VLAN Cons:

- Adds complexity and cost (needs Layer 3 device)
- Can introduce latency if not optimized

4. Applications of VLAN and Inter-VLAN

- VLANs are used in offices, schools, and data centers to separate departments (e.g., HR, Finance, IT).
- **Inter-VLAN** routing is used when these departments need to securely share resources like servers or printers across VLANs.

5. What is Trunking and dot1q Encapsulation?

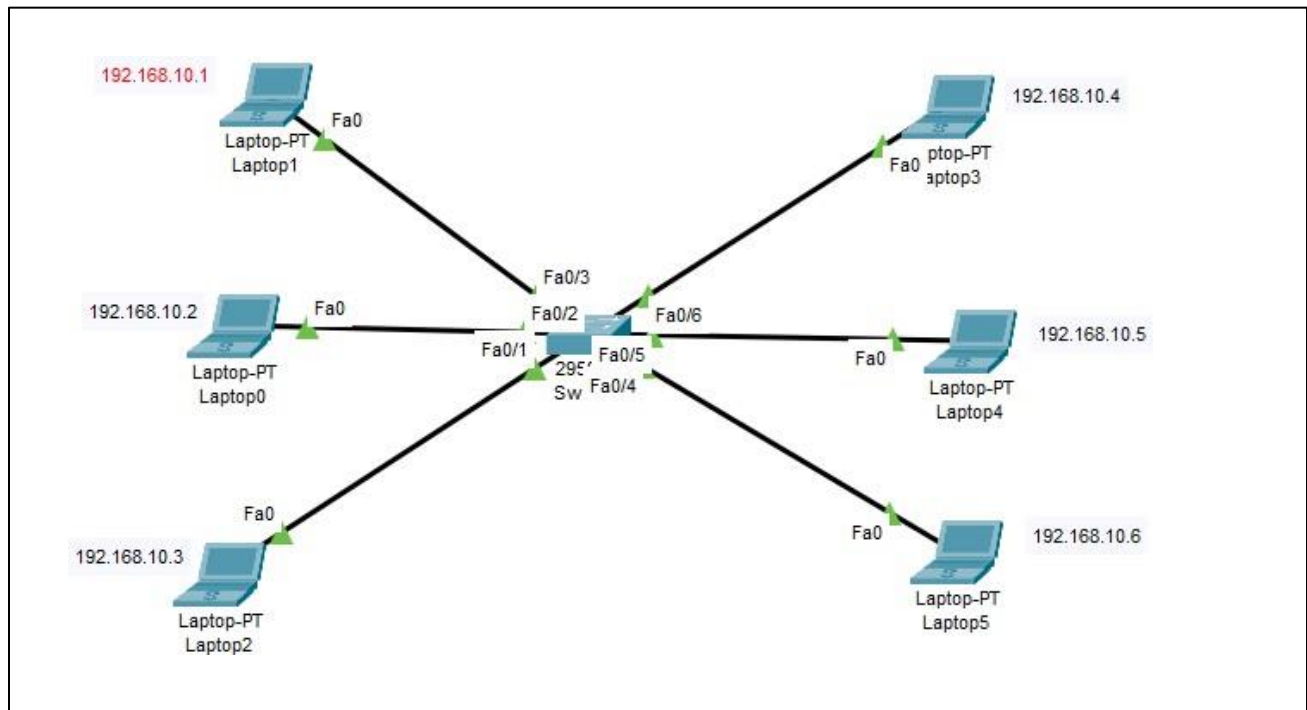
Trunking allows multiple VLANs to travel over a single physical link between switches or

routers.

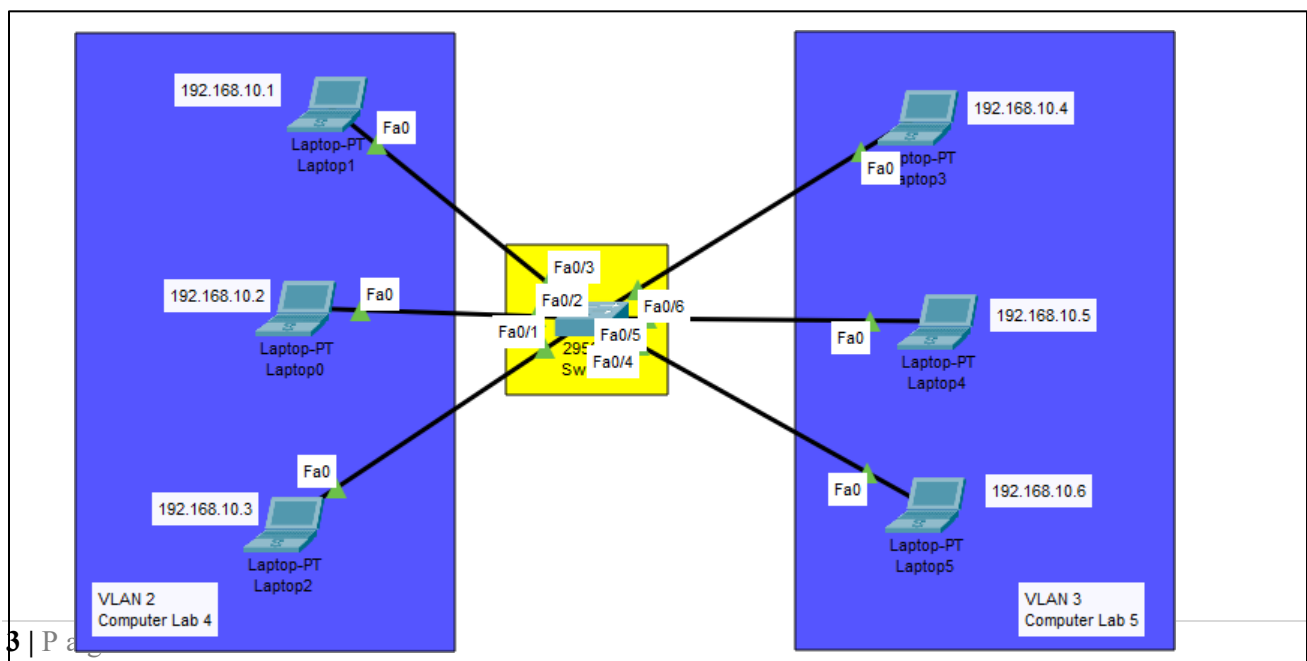
Dot1Q (802.1Q) is a tagging method used in trunking to identify VLANs by adding a tag to Ethernet frames, enabling proper delivery to the correct VLAN.

TASK # 01

- Assigning Ips:**



- Highlight Networks:**



▪ Vlan Configuration

```
Switch(config)#Vlan 2
Switch(config-vlan)#Name cslab4
Switch(config-vlan)#Vlan 5
Switch(config-vlan)#Name cslab5
Switch(config-vlan)#Exit
Switch(config)#Int fa0/1
Switch(config-if)#Switchport access vlan 2
Switch(config-if)#Exit
Switch(config)#Int fa0/4
Switch(config-if)#Switchport access vlan 5
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

• Showing Vlan Brief:

```
Switch#show vlan brief
```

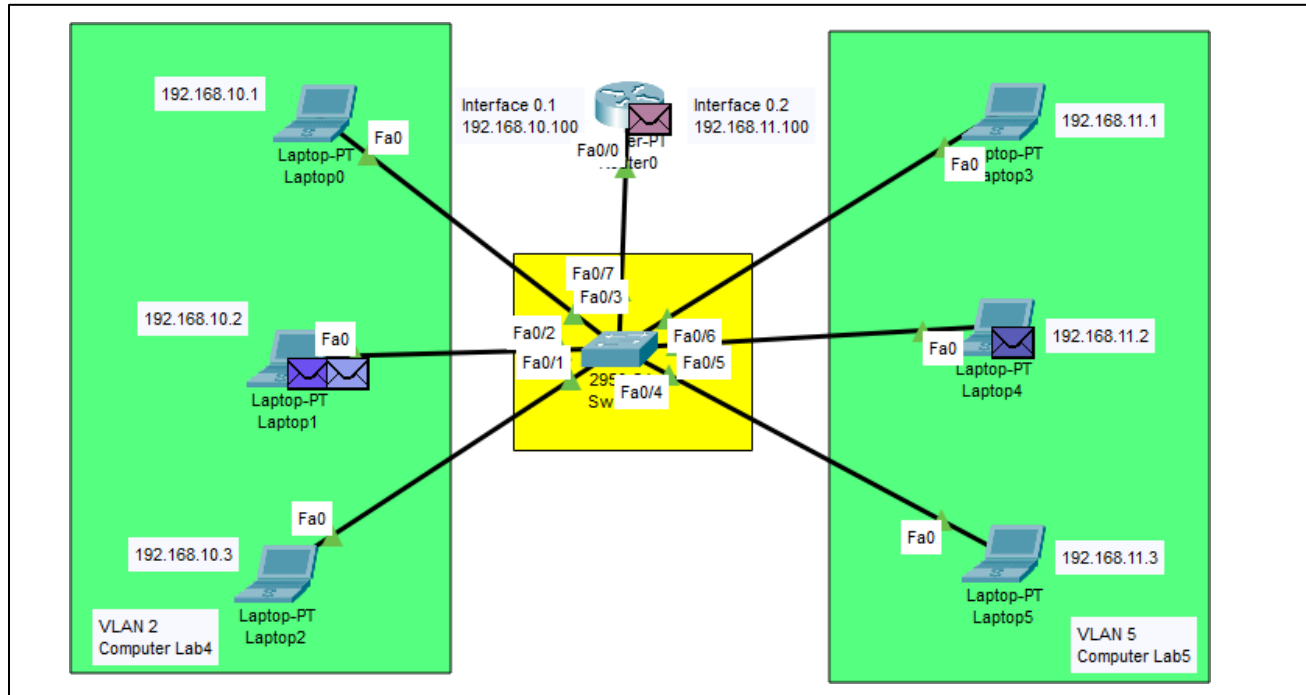
VLAN	Name	Status	Ports
1	default	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24
2	cslab4	active	
5	cslab5	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Realtime Simulation									
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	Laptop2	Laptop1	ICMP		0.000	N	0	(ec
	Successful	Laptop0	Laptop1	ICMP		0.000	N	1	(ec
	Successful	Laptop3	Laptop0	ICMP		0.000	N	2	(ec

TASK # 02

■ Assigning Ips



■ Switch Configuration

```
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/7
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#
```

■ Router Configuration

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 unassigned      YES unset    administratively down down
FastEthernet1/0 unassigned      YES unset    administratively down down
Serial2/0       unassigned      YES unset    administratively down down
Serial3/0       unassigned      YES unset    administratively down down
FastEthernet4/0 unassigned      YES unset    administratively down down
FastEthernet5/0 unassigned      YES unset    administratively down down
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no shutdown
```

```

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	up	up
FastEthernet1/0	unassigned	YES	unset	administratively down	down
Serial2/0	unassigned	YES	unset	administratively down	down
Serial3/0	unassigned	YES	unset	administratively down	down
FastEthernet4/0	unassigned	YES	unset	administratively down	down
FastEthernet5/0	unassigned	YES	unset	administratively down	down

```

Router#

```

▪ Creation of sub Interface and route

```

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0.1
Router(config-subif)#Encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.10.100 255.255.255.0
Router(config-subif)#exit
Router(config)#int fa0/0.2
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.2, changed state to up

```

```

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up
Encapsulation dot1Q 5
Router(config-subif)#ip address 192.168.11.100 255.255.255.0
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

```

show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	up	up
FastEthernet0/0.1	192.168.10.100	YES	manual	up	up
FastEthernet0/0.2	192.168.11.100	YES	manual	up	up
FastEthernet1/0	unassigned	YES	unset	administratively down	down
Serial2/0	unassigned	YES	unset	administratively down	down
Serial3/0	unassigned	YES	unset	administratively down	down
FastEthernet4/0	unassigned	YES	unset	administratively down	down
FastEthernet5/0	unassigned	YES	unset	administratively down	down

```

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#Network 192.168.10.0
Router(config-router)#Network 192.168.11.0
Router(config-router)#exit
Router(config)#

```

- Assign interface 0/0.1 ip address as default gateway to responding Vlan end User Devices and so on.

✓ Vlan 2 and 5:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	Laptop0	Laptop2	ICMP		0.000	N	0	(edit)
	Successful	Laptop3	Laptop5	ICMP		0.000	N	1	(edit)

✓ Ping router subinterfaces from each PC:

```
C:\>ping 192.168.10.100
```

```
Pinging 192.168.10.100 with 32 bytes of data:
```

```
Reply from 192.168.10.100: bytes=32 time=4ms TTL=255
Reply from 192.168.10.100: bytes=32 time=4ms TTL=255
Reply from 192.168.10.100: bytes=32 time=4ms TTL=255
Reply from 192.168.10.100: bytes=32 time=4ms TTL=255
```

From VLAN2 PC

```
Ping statistics for 192.168.10.100:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms
```

```
C:\>ping 192.168.10.100
```

```
Pinging 192.168.10.100 with 32 bytes of data:
```

```
Reply from 192.168.10.100: bytes=32 time=4ms TTL=255
Reply from 192.168.10.100: bytes=32 time=4ms TTL=255
Reply from 192.168.10.100: bytes=32 time=4ms TTL=255
Reply from 192.168.10.100: bytes=32 time=4ms TTL=255
```

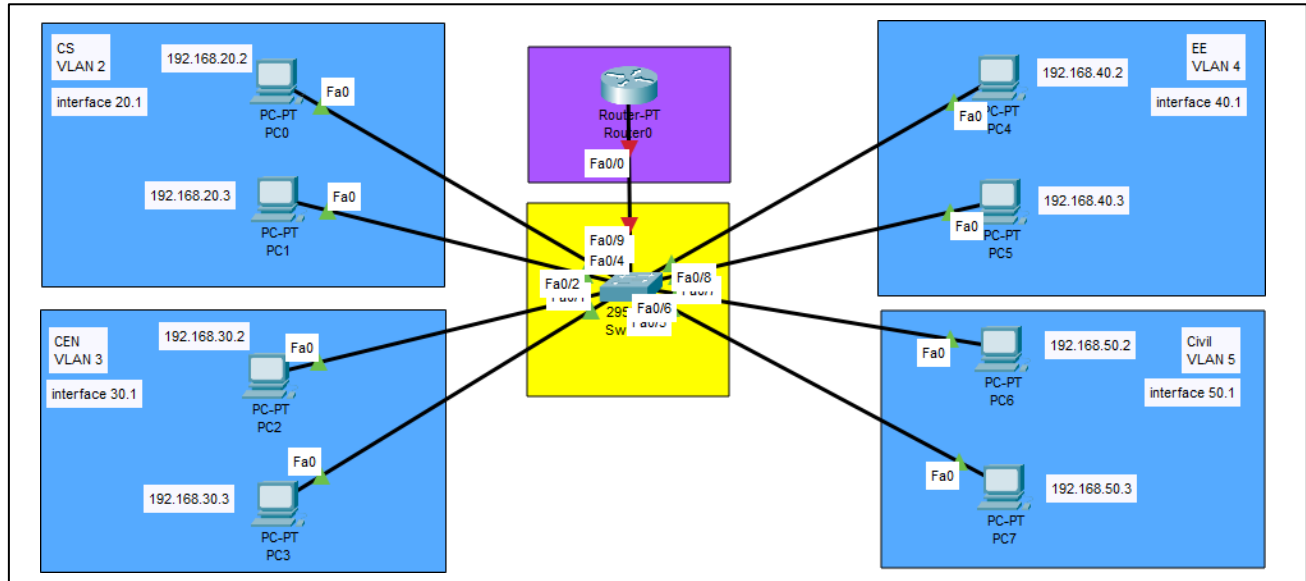
From VLAN5 PC

```
Ping statistics for 192.168.10.100:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms
```

TASK # 03

- Label each device with unique IP address.



- Pc IP configuration:

IPv4 Address	192.168.20.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1
DNS Server	0.0.0.0

- Configure switch for Vlan

```
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name CS
Switch(config-vlan)#vlan 3
Switch(config-vlan)#name CEN
Switch(config-vlan)#vlan 4
Switch(config-vlan)#name EE
Switch(config-vlan)#vlan 5
Switch(config-vlan)#civil
Switch(config-vlan)#^
% Invalid input detected at '^' marker.
Switch(config-vlan)#vlan 5
Switch(config-vlan)#name civil
```

▪ Configuring Access ports:

```
Switch(config-vlan)#interface range Fa0/2 - 3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 2
Switch(config-if-range)#interface range Fa0/6 - 7
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 3
Switch(config-if-range)#interface range Fa0/8, Fa0/10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 4
Switch(config-if-range)#interface range Fa0/5, Fa0/9
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 5
```

▪ Configuring Trunk port:

```
Switch(config-if-range)#interface Fa0/4
Switch(config-if)#switchport mode trunk
```

VLAN Name	Status	Ports
1 default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24
2 CS	active	Fa0/2, Fa0/3
3 CEN	active	Fa0/6, Fa0/7
4 EE	active	Fa0/8, Fa0/10
5 civil	active	Fa0/5, Fa0/9
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Name: Fa0/4
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
```

Router configuration for Vlan

```
Router>en
Router#Show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          unassigned      YES unset    administratively down down
FastEthernet1/0          unassigned      YES unset    administratively down down
Serial2/0                 unassigned      YES unset    administratively down down
Serial3/0                 unassigned      YES unset    administratively down down
FastEthernet4/0          unassigned      YES unset    administratively down down
FastEthernet5/0          unassigned      YES unset    administratively down down
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          unassigned      YES unset    up          up
FastEthernet1/0          unassigned      YES unset    administratively down down
Serial2/0                 unassigned      YES unset    administratively down down
Serial3/0                 unassigned      YES unset    administratively down down
FastEthernet4/0          unassigned      YES unset    administratively down down
FastEthernet5/0          unassigned      YES unset    administratively down down
```

Router sub Interface Configuration

```
interface FastEthernet0/0
  no ip address
  duplex auto
  speed auto
!
interface FastEthernet0/0.2
  encapsulation dot1Q 2
  ip address 192.168.20.1 255.255.255.0
!
interface FastEthernet0/0.3
  encapsulation dot1Q 3
  ip address 192.168.30.1 255.255.255.0
!
interface FastEthernet0/0.4
  encapsulation dot1Q 4
  ip address 192.168.40.1 255.255.255.0
!
interface FastEthernet0/0.5
  encapsulation dot1Q 5
  ip address 192.168.50.1 255.255.255.0
!
```